Number Theory 1 – G11

Theory

Division remainders

If a is a divisor of b, we write $a \mid b$ (read: a divides b). Thus:

$$a \mid b \text{ if and only if there exists an integer } c \text{ such that } \quad b = ac.$$

If a is not a divisor of b, we write $a \nmid b$ (read: a does not divide b).

Suppose we have integers a and b with $b > 0$. We say that a gives quotient q and remainder r when divided by b if

$$a = b \cdot q + r \quad \text{and} \quad 0 \leqslant r < b.$$

Claim 1. If a, b are positive and $a \mid b$, then $a \leqslant b$.

Claim 2. The only number divisible by all natural numbers is 0.

Remark 1 (Trick for representing nubers with repeating digits)**.** How to represent a number with n identical digits in compact form? Note that $10^n - 1$ is a number consisting of n nines, so:

$$\underbrace{kkk\ldots k}_{n \text{ digits } k} = \frac{10^n - 1}{9} \cdot k.$$

Congruences

Suppose we have a positive integer n. We say that two integers a and b are congruent modulo n if and only if a and b give the same remainder when divided by n. We write $a \equiv b \pmod{n}$. Equivalently:

$$a \equiv b \pmod{n} \text{ if and only if } \; n \mid a - b.$$

Addition, multiplication, and subtraction modulo n behave as expected:

$$a \pmod{n} + b \pmod{n} \quad \equiv_n \quad (a+b) \pmod{n},$$
$$a \pmod{n} \cdot b \pmod{n} \quad \equiv_n \quad ab \pmod{n}.$$

Division modulo n is generally only defined when n is prime. Of course, division by 0 is still undefined, meaning we cannot divide by numbers divisible by n.

The inverse of a number a modulo n is a number b such that $ab \equiv_n 1$. This b exists if and only if $a \perp n$ and is denoted a^{-1} or $\frac{1}{a}$.

Prime numbers

Definition 1. A natural number different from 1 whose only divisors are 1 and itself is called a **prime number**.

Every number can be represented as a product of prime numbers (the uniqueness of this factorization can be proved using the following lemmas). There are infinitely many primes, which can be proved by considering $p_1 \cdot p_2 \cdot \dots \cdot p_k + 1$.

Lemma 1. Prime numbers greater than 3 are congruent to 1 or 5 mod 6.

About GCD

Definition 2 (Greatest Common Divisor). We say $d = \gcd(a, b)$ if $d \mid a$ and $d \mid b$ (i.e., it is a common divisor) and for every common divisor e of a and b, $e|d$ (i.e., every common divisor divides the gcd).

$\gcd(a, b)$ is the smallest positive number of the form $ax + by$ for integers x, y (in other words, it is their smallest linear combination). It can be shown that this definition gives the same gcd as the standard definition.

Definition 3. Natural numbers a and b are **coprime** if $\gcd(a, b) = 1$, denoted $a \perp b$.

Lemmas (notation: p is prime, n, a, b are integers):

- If $a \perp n$ and $n \mid ab$ then $n \mid b$.
- If $p \nmid a$ then $a \perp p$.
- If $p \mid ab$ then $p \mid a$ or $p \mid b$.

Lemma 2. Given a permutation σ_i of numbers from 0 to $n-1$ and a number p coprime with n ($p \perp n$), after multiplying each element of the permutation by p ($\sigma_i' = p \cdot \sigma_i$) and taking the remainder modulo n, we obtain a new permutation of numbers from 0 to $n-1$.

Proof. It suffices to show that each number maps to a different position. If not, there would exist two numbers $\sigma_i, \sigma_j < n$ such that:

$$p \cdot (\sigma_i - \sigma_j) \equiv 0 \pmod{n}$$

Since $p \perp n$, we have $\sigma_i - \sigma_j \equiv 0 \pmod{n}$, so $\sigma_i = \sigma_j$. ∎

Theorem 1 (Euclidean Algorithm). We can find gcd without factoring numbers into primes (in particular, without knowing what primes are) by repeatedly using the relation $\gcd(a, b) = \gcd(a - kb, b)$. By reversing this algorithm, we can express gcd as a linear combination of a and b.

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Important theorems

Theorem 2 (Fermat's Little Theorem). For any natural number n and any prime p:

$$p \mid n^p - n,$$

or in congruence form:

$$n^p \equiv n \pmod{p}.$$

Proof. Divide a circle into p parts and consider colorings with n different colors.
There are n^p colorings in total, and n monochromatic colorings (all parts same color).
The remaining colorings can be grouped into sets of p - if a coloring can be obtained by rotating another, they belong to the same group.

We show each group has exactly p colorings: assume by contradiction that after rotating some coloring by k segments ($0 \leqslant k < n$), we get the same coloring. Since p is prime, $p \perp k$, so for a segment to return to its initial position, we must perform p rotations, meaning all segments have the same color - contradiction. ∎

Theorem 3 (Wilson's Theorem). p is prime if and only if:

$$(p-1)! \equiv -1 \pmod{p}$$

Proof. Consider the equation $x^2 - 1 = 0$ in $\mathbb{Z}_p$. Solutions are numbers that are their own inverses. We have $x^2 - 1 = (x-1)(x+1)$, so there are only two solutions: $x = 1$ and $x = -1$, where $-1 \equiv p - 1$. Thus, all remaining numbers are in the set $\{2, \dots, p-2\}$ and have their inverses in this set. We can pair each with its inverse, so multiplying them gives

$$2 \cdot 3 \cdot \ldots \cdot (p-2) \equiv 1 \pmod{p}$$

Multiplying by 1 and $p - 1$ yields the result. ∎

Theorem 4 (Chinese Remainder Theorem). Let $m_1, m_2, ..., m_k$ be pairwise coprime and $a_1, a_2, ..., a_k$ be integers. Then the system:

$$\begin{cases} x \equiv a_1 \pmod{m_1} \\ x \equiv a_2 \pmod{m_2} \\ \vdots \\ x \equiv a_k \pmod{m_k} \end{cases}$$

is equivalent to a single congruence $x \equiv A \pmod{m_1 m_2 ... m_k}$.

Technique: **Infinite Descent**
This method involves taking the smallest solution (in a chosen context) and showing there exists a smaller one.

Example 1. Find all positive integer solutions to

$$x^3 + 2y^3 = 4z^3$$

Solution 1. Take a solution where $x + y + z$ is minimized. Since:

$$2y^3 \equiv 0 \pmod 2,$$
$$4z^3 \equiv 0 \pmod 2,$$

x must be divisible by 2.
Substitute $x' = x/2$ and divide by 2:

$$4x'^3 + y^3 = 2z^3$$

Repeat twice more to get the original equation.
Contradiction since x, y, z were minimal. Thus no solutions exist.

Example 2. Find all non-negative integers a, b such that $ab \mid a^n + b$, where $n \in \mathbb{N}_{>1}$ is fixed.

Solution 2. Note a and b have the same divisors, and $a = 1$ iff $b = 1$. Assume $a, b > 1$ and take a pair with minimal sum. Let $p \in \mathbb{P}$ be a prime dividing both a and b. Then $p^2 \mid a^n + b$, implying $p^2 \mid b$. Repeat until $p^n \mid b$. Let $a = pa_1$, $b = p^n b_1$. Then:

$$ab = a_1 b_1 p^{n+1} \qquad a^n + b = p^n(a_1^n + b_1).$$

Thus:

$$p^{n+1} a_1 b_1 \mid p^n(a_1^n + b_1) \implies pa_1 b_1 \mid a_1^n + b_1.$$

If $a_1 \geqslant 2$ or $b_1 \geqslant 2$, then we could construct a new solution with a smaller sum, contradicting minimality (because $p \geqslant 2$ and $n \geqslant 1$, so $p^n \geqslant 4$ and $a_1 + b_1 \geqslant 2$):

$$a_1 + b_1 < pa_1 + p^n b_1 = a + b.$$

Thus $a_1 = b_1 = 1$, so $a = p$, $b = p^n$. Substituting back gives:

$$p^{n+1} \mid 2p^n \implies p = 2.$$

So the solution is $a = 2$, $b = 2^n$.

p-adic valuations

- Prime factorization is a powerful way to look at numbers. p-adic valuations will help us with this. For prime p, the p-adic valuation of n is the exponent of p in the prime factorization of n, denoted $v_p(n)$.
- $p^{v_p(n)} \mid n$ but $p^{v_p(n)+1} \nmid n$.
- $a \mid b$ iff for every prime p, $v_p(a) \leqslant v_p(b)$
- $a = b$ iff for every prime p, $v_p(a) = v_p(b)$
- $v_p(ab) = v_p(a) + v_p(b)$
- $v_p(\frac{a}{b}) = v_p(a) - v_p(b)$. This extends p-adic valuations to rational numbers.
- When $v_p(a) \neq v_p(b)$, $v_p(a + b) = \min(v_p(a), v_p(b))$.

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Warm-up

1. Prove the following properties of congruences.
   - If $a \equiv b \pmod{m}$, then $b \equiv a \pmod{m}$.
   - If $a \equiv b \pmod{m}$ and $b \equiv c \pmod{m}$, then $a \equiv c \pmod{m}$.
   - $a \equiv b \pmod{m}$ iff $ac \equiv bc \pmod{mc}$.
   - If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then $a + c \equiv b + d \pmod{m}$, $a - c \equiv b - d \pmod{m}$, $ac \equiv bd \pmod{m}$.
2. Show that if $p \nmid n$, then $n^{p-1} \equiv 1 \pmod{p}$.
3. Express $\underbrace{222\ldots2}_{2024}\underbrace{444\ldots4}_{2024}$ as a sum (using the trick).

Problems

1. For which natural n is the fraction $\frac{21n+4}{14n+3}$ irreducible?
2. Find the last digit of $2023^{2024^{2025}}$.
3. Prove $F_n \perp F_{n+1}$ for all natural n, where F_n is the n-th Fibonacci number ($F_0 = 0$, $F_1 = 1$, $F_n = F_{n-1} + F_{n-2}$ for $n > 1$).
4. Do there exist 1000 consecutive natural numbers such that exactly 5 are prime?
5. Is $4^6 + 4 \cdot 6^5 + 9^5$ composite?
6. Find divisibility rules for 11 and 101. Find a divisibility rule for 7 using $7 \mid 1001$.
7. Select $n + 1$ numbers from $\{1, 2, 3, \ldots 2n\}$. Prove that some two are coprime.
8. Prove there are infinitely many primes congruent to 3 mod 4. Does the same proof work for primes of the form $4k + 1$?
9. Prove that for natural n and odd k:
$$1 + 2 + 3 + \cdots + n \mid 1^k + 2^k + 3^k + \cdots + n^k$$
10. *VII Polish MO* Prove $2x^2 - 215y^2 = 1$ has no integer solutions.
11. Prove $120 \mid n^5 - 5n^3 + 4n$ for all integers n.
12. Find all integers k such that $n \mid (n-1)^k + 1$ for all natural n.
13. Prove there are infinitely many primes of the form $\sqrt{24n+1}$ for some natural n.
14. Does there exist an infinite non-constant arithmetic progression consisting entirely of primes?
15. A right triangle has legs a, b and hypotenuse c. Prove that if a, b, c are integers, at least one of a, b is even.

16. Find the number of divisors of $16 \cdot 27 \cdot 49 \cdot 19$. Find their sum.
17. Prove that if a, b, c, d, e satisfy $9 \mid a^3 + b^3 + c^3 + d^3 + e^3$, then $3 \mid abcde$.
18. Prove that for any natural numbers $a_1, a_2, ..., a_n$ with $\gcd(a_1, a_2, ..., a_n) = 1$, there exist integers $k_1, k_2, ..., k_n$ such that
$$k_1a_1 + k_2a_2 + ... + k_na_n = 1$$
19. Let a be any natural number. Prove that among $a, a+1, a+2, ..., a+9$, there is one coprime to all others.
20. Given $v_p(a), v_p(b)$, compute $v_p(\gcd(a, b))$ and $v_p(\mathrm{lcm}(a, b))$.
21. Prove:
$$\frac{\mathrm{lcm}(a,b)\mathrm{lcm}(b,c)\mathrm{lcm}(c,a)}{\mathrm{lcm}(a,b,c)^2} = \frac{\gcd(a,b)\gcd(b,c)\gcd(c,a)}{\gcd(a,b,c)^2}$$
22. Given integers a, b such that $\frac{a^2}{b} + \frac{b^2}{a}$ is integer. Prove that $\frac{a^2}{b}$ and $\frac{b^2}{a}$ are integers.
23. Prove that for natural m and n:
$$\gcd(a^m - 1,\ a^n - 1) = a^{\gcd(m,n)} - 1$$
24. Prove $2^{2^n} - 1$ has at least n distinct prime divisors for all natural n.
25. *II Polish MO* Prove $7 \mid 13^n + 6$ for even natural n.
26. *IV Polish MO* Prove $11 \mid 2^{55} + 1$.
27. Find the remainder when $3^{81} + 7^{72}$ is divided by 11.
28. Prove the last digit of 7^{256} is 1.
29. Prove $7 \mid 2222^{5555} + 5555^{2222}$.
30. Find the last two digits of 2^{999}.
31. Prove $29 \mid 2^{5n+1} + 3^{n+3}$ for all natural n.
32. *VI Polish MO* Find the last digit of $53^{53} - 33^{33}$.
33. Show $1\underbrace{000\ldots0}_{2013}1$ is composite.
34. Prove $F_n \mid F_{nk}$ for all natural n, k.
35. Prove distinct Fermat numbers $G_n = 2^{2^n} + 1$ are pairwise coprime.
36. Find a formula for $v_p(n!)$
37. Natural numbers a, b satisfy: For all natural n, $a^n \mid b^{n+1}$. Prove $a \mid b$.
38. Prove $(n!)^{(n-1)!} \mid (n!)!$ for all natural n.
39. (Polish MO nr. 75, stage 2) Let p be prime. Prove $p(p^2 \cdot \frac{p^{p-1}-1}{p-1})!$ is divisible by $p! \cdot (p^2)! \cdot (p^3)! \cdot \ldots \cdot (p^p)!$.